

# Global Drone Regulatory Landscape

## From Fragmentation to Risk-Based Harmonization

An Advanced Comparative Analysis of Sovereign Airspace Frameworks,  
UTM Architectures, and BVLOS Integration.

telemetry readout

[SYS.READY]

Global Drone Economy Valuation: \$63  
Billion (by 2025 - IATA)

**DOMAINS:** FAA | EASA | UK CAA | CAAC | ICAO



# The Regulatory Imperative: Reconciling Conflicting Forces

## The Economic Imperative (Global Standardization)



- Driven by cross-border logistics and humanitarian missions.
- Demands ICAO interoperability guidelines to scale investments and address operational uncertainties.
- **Key Stat:** Harmonization is required to unlock the projected \$63B global economy, shifting ad-hoc flights to financier-attractive, scalable operations.

## The Sovereign Imperative (Local Oversight)



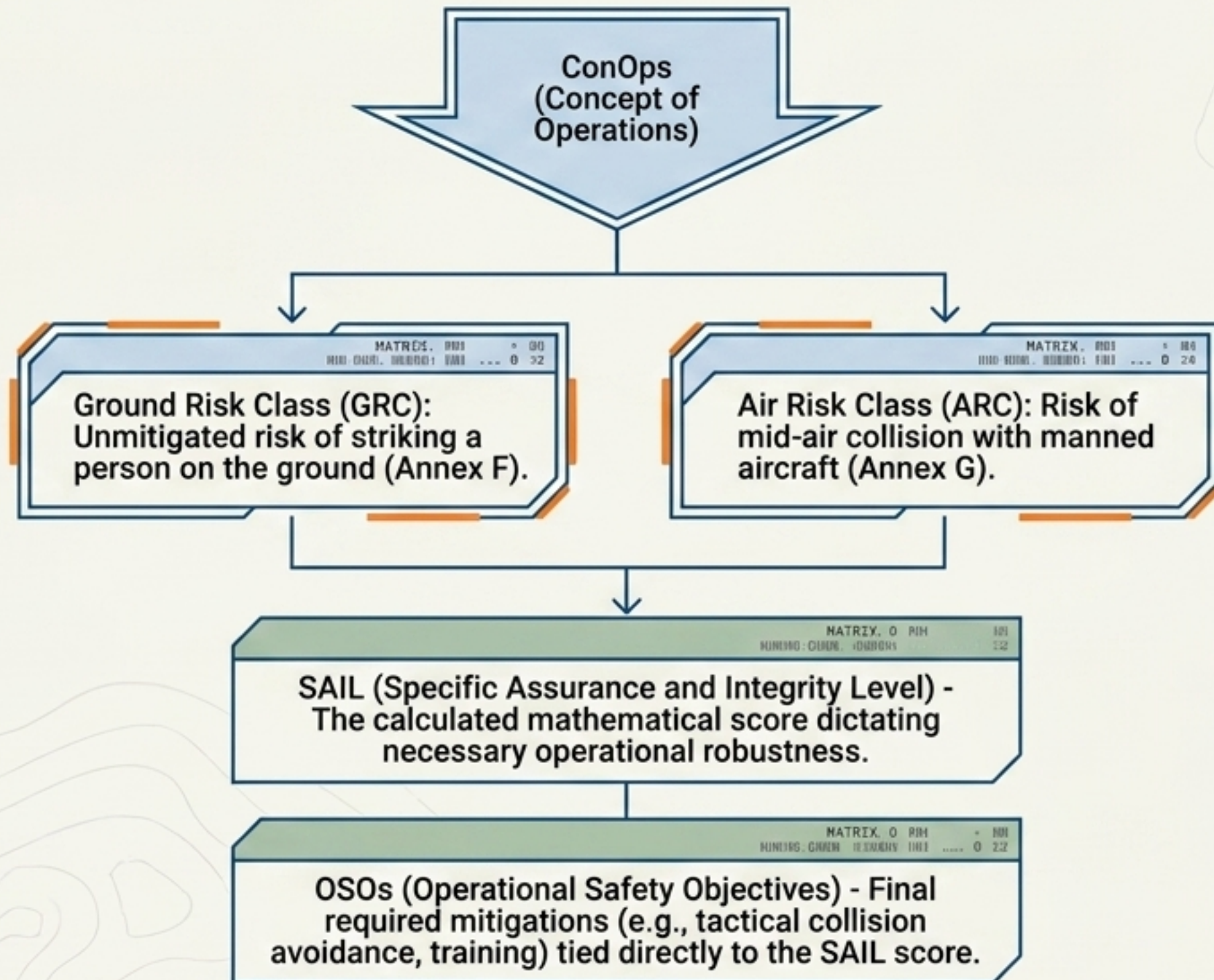
- Driven by national defense, privacy concerns, and targeted airspace control.
- Requires localized restrictions for critical infrastructure.
- **Example:** Germany and France maintaining strict sovereign oversight and distinct rules near sensitive borders.

## The Structural Solution

The only way to reconcile these opposing forces is by replacing rigid, rule-based regimes with dynamic, performance-based risk methodologies.



# The Risk Engine: Unpacking the JARUS SORA Methodology



## Global Anchor

JARUS developed this independent of ICAO. It provides the gold standard that EASA adopted as an Acceptable Means of Compliance for Regulation (EU) 2019/947 (Specific Category).



# EASA "Open Category" Diagnostic Table

| Subcategory             | Hardware<br>(Class Marks)                          | Operational Limits & Distance                                                                                                     |
|-------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| A1<br>(Over People)     | Class 0 (<250g)<br>and Class 1                     | Minimize flight over uninvolved people; no assemblies. No remote pilot training required for C0.                                  |
| A2<br>(Close to People) | Class 2                                            | Must maintain a lateral distance equal to flight height (1:1 rule). Never closer than 30m horizontally (or 5m in low-speed mode). |
| A3<br>(Far from People) | Class 3, Class 4,<br>Privately built up<br>to 25kg | Must operate 150m from residential, commercial, or industrial areas. Zero uninvolved people present.                              |

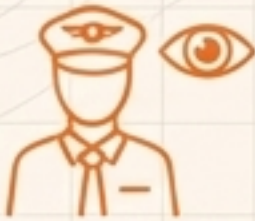
## EASA Framework Mapping:

- OPEN:  
Low Risk
- SPECIFIC:  
Medium Risk  
(Requires SORA / PDRA / STS / LUC)
- CERTIFIED:  
High Risk

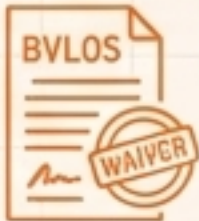


# US FAA: Moving from Waivers to Performance-Based Scalability

## Legacy System: Part 107



- **Pilot-centric** model requiring strict visual line of sight.



- BVLOS requires individual, ad-hoc waivers demonstrating sufficient mitigations.

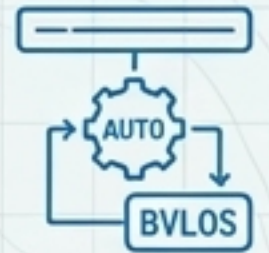


- **Operational Bottleneck:** A pipeline operator must juggle 20 separate BVLOS waivers just to keep routine inspections running.

Architectural Shift to  
Advanced Autonomy

## Scalable Future: Parts 108 & 146

- **Part 108 (Operations):** Standardized framework for routine BVLOS and automated systems (including >55 lbs), eliminating routine waivers.



- **Liability Shift:** Focus moves from individual pilots to an Operations Supervisor. Primary safety liability shifts to the corporate operator.



- **Part 146 (Infrastructure):** Certifies and oversees Automated Data Providers—third-party UTM services handling safety functions like drone deconfliction.





# UK CAA: Post-Brexit Independence & Domestic Tailoring

## [SYS.01] Phasing out PfCO for UK SORA



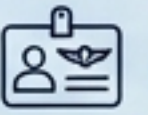
- The UK has officially phased out the legacy Permission for Commercial Operations (PfCO).
- Transitioning from the older Operating Safety Case (OSC) method to the new UK SORA methodology, effective April 2025.
- Allows highly tailored requirements for domestic use cases like swarms or cargo delivery.

## [SYS.02] Equipment Standards



- Legacy EU C-class drones will be accepted in the UK until December 31, 2027.
- **Hard Deadline:** All new models placed on the market from January 1, 2026, must carry explicit UK class marks (UK0 to UK6).

## [SYS.03] Personnel Identification



- The UK enforces a mandatory dual-ID registration system.
- Operator ID: Must be physically displayed on the body of the drone.
- **Flyer ID:** Requires a formal competency test for operating any drone weighing more than 250g.

// NOTE: CAA framework aligns with international best practices while localizing hardware validation.



# Regulatory Philosophies: Process Oversight vs. Hardware Control

## Western Software & Process Model (FAA/EASA)



- **Mathematical Risk Focus:** Relies heavily on mathematical operational risk assessments (like SORA).



- **Software Ecosystem:** Relies on automated data providers (Part 146) and third-party UTM systems for dynamic airspace management.



- **Pilot & Operator Identity:** Software-managed registration and remote pilot certification.

## China CAAC Hardware Model



- **Identification:** Real-name registration required for all drones, regardless of weight, tied directly to a Chinese mobile number and WeChat account.



- **Hardware Mandates (2026):** Drones legally classified as aircraft. Must be physically incapable of flight before activation and after deactivation. Every unit requires a built-in operational ID module.



- **Airspace Lockout:** Hardcoded 120m AGL maximum altitude limit. Total hardware ban on drone usage in Beijing (effective May 2026, including transport via train/flight) and Shanghai.

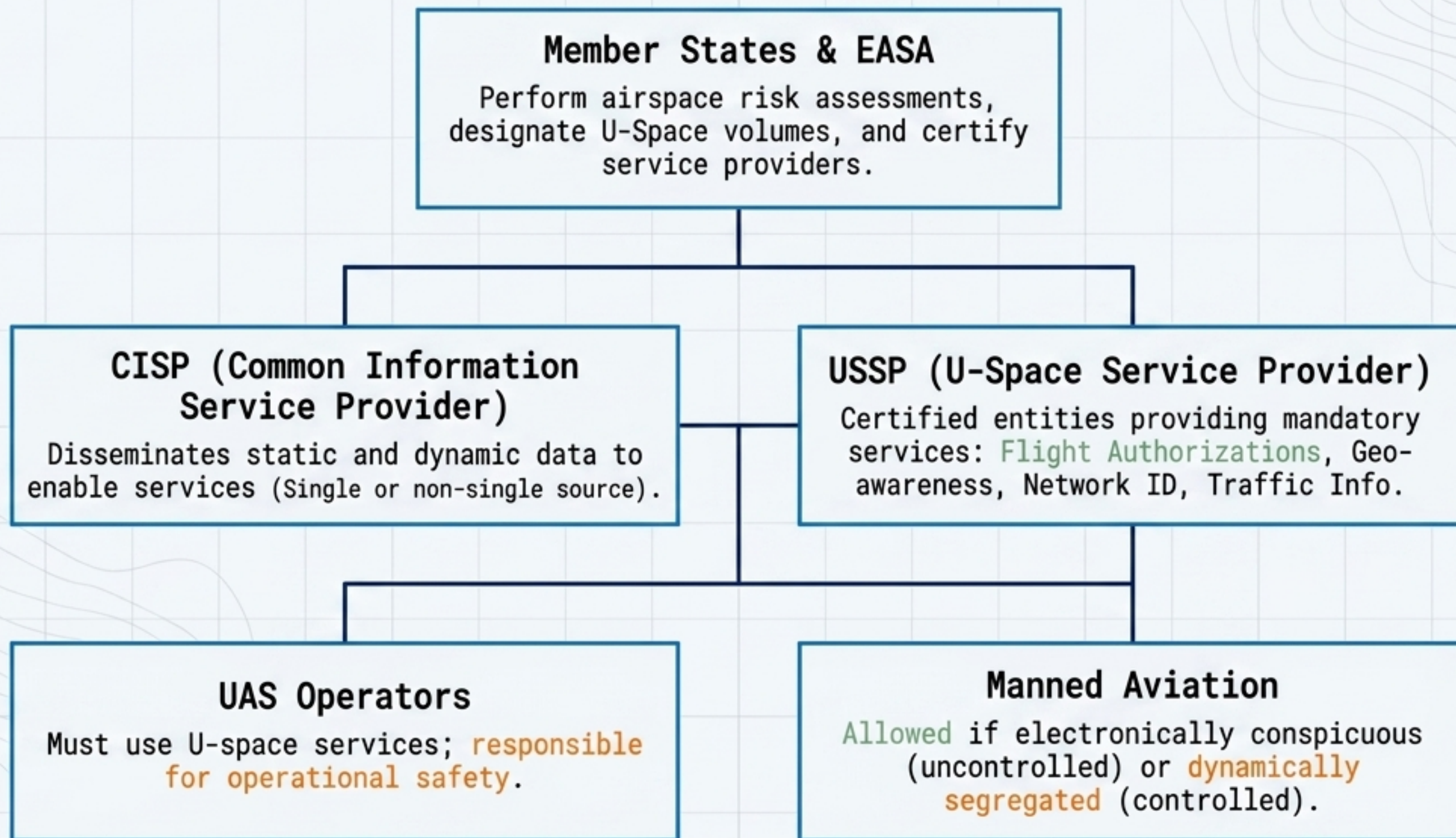


# BVLOS Comparative Matrix: The Ultimate Operational Challenge

| Diagnostic Criteria           | United States (FAA)                                                                           | European Union (EASA)                                  | United Kingdom (CAA)                                                       |
|-------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------------|
| Primary Framework             | Part 108 & Part 146 (Rulemaking)                                                              | Specific Category / JARUS SORA                         | UK SORA (Post-OSC, April 2025)                                             |
| Primary Liability & Personnel | Corporate Operator / Operations Supervisor (No traditional airman cert for automated flights) | Remote Pilot / Operator                                | Remote Pilot / Operator                                                    |
| Traffic Management Strategy   | Certified Automated Data Providers (Part 146)                                                 | U-Space Airspace / USSPs                               | Implementation of long-range BVLOS in Atypical environments by end of 2025 |
| Airworthiness Approach        | Reliance on consensus industry standards and real-world testing                               | Design Verification Reports (DVR) / EASA Certification | CAA validation and domestic equipment standards                            |



# U-Space Network Architecture Diagram



U-SPACE EVOLUTION PHASES: U1 (Foundation) -> U2 (Initial) -> U3 (Advanced/Automated DAA) -> U4 (Full automation/connectivity)



# The Future of Convergence (2026+)

## 1. The Ubiquity of Electronic Conspicuity

Remote ID and network identification are becoming non-negotiable baselines across the FAA, EASA, and CAAC. EASA, and CAAC. This effectively digitizes all low-altitude traffic, creating a foundational layer for autonomous tracking.

## 2. The Shift from Hardware to UTM Software

Scaling BVLOS will rely less on individual drone hardware capabilities and entirely on the certification of third-party network managers, such as USSPs in Europe and Automated Data Providers via Part 146 in the US.

## 3. Standardization Triumphs

Adoption of Standard Scenarios (STS) and Predefined Risk Assessments (PDRA) will commoditize routine urban flights. ICAO's global UAS interoperability guidelines will force sovereign networks to communicate, enabling cross-border autonomous logistics.